FLAVIO TOFFALINI

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My research interest covers many aspects of system security. My Ph.D. background focuses on software security for Trusted Execution Environments.

CURRENT POSITION: ASSISTANT PROFESSOR

Ruhr-Universität Bochum (RUB), Germany *Sep 2024 to Now*
Assistant Professor in Automated Security Analysis (fixed-term ending 2030)

FORMER POSITION

École Polytechnique Fédérale de Lausanne (EPFL), Switzerland *Nov 2021 – Aug 2024*
PostDoc, supervised by Prof. Mathias Payer
Topic: fuzzing, mitigation, software analysis

EDUCATION

Singapore University of Technology and Design, Singapore *Jan 2017 - Sep 2021*
Ph.D., Supervisor Prof. Jianying Zhou
Topic: trusted computing, system security
Thesis Title: Challenges, threats, and novel defenses for Trusted Execution Environments

University of Verona, Italy *Sep 2012 - Oct 2015*
M.S. in Computer Science and Engineering 108/110, GPA 3,9/4
Supervisor Prof. Damiano Carra
Master thesis topic: Google dorks, Web security

University of Pavia, Italy *Sep 2006 - Dec 2009*
B.S. in Computer Engineering 101/110, GPA 3,67/4
Supervisor Prof. Paolo Gamba

ACADEMIC ACTIVITIES

King's College London Nov 2019 - Mar 2020
Visiting fellow, supervised by Prof. Lorenzo Cavallaro *London, UK*
Topic: trusted computing, system security

University of Padua Jan 2018 - Aug 2018
Visiting Fellow, supervised by Prof. Mauro Conti *Padua, Italy*
Topic: trusted computing, system security

University of Verona Dec 2015 - July 2016
Research Assistant, supervised by Prof. Fausto Spoto *Verona, Italy*
Topic: static analysis of Android applications

Eurecom April 2015 - July 2015
Visiting Fellow, supervised by Prof. Davide Balzarotti *Biot, France*
Topic: Google dorks, Web security

TEACHING ACTIVITIES

"Advanced Automatic Testing" (Master Course, Main Instructor) at Ruhr-Universität Bochum, 48 hours

"Seminar in Advanced Automatic Testing" (Master/Bachelor Seminar, Main Instructor) at Ruhr-Universität Bochum, 48 hours

"Programming and programming languages" (Bachelor Course, 2nd semester, Co-Instructor 30%) at Ruhr-Universität Bochum, 48 hours

"Securing Digital Transformation: Approaches and Technologies" (SERICS PhD School, Invited Lecturer), 2025

"Security Testing and Verification" (Summer School, Invited Lecturer), 2026

SUPERVISION

PhDs: Currently 2, 2 co-supervised during Post-Doctoral

RA: Currently 1, 1 successfully completed

MSc: Currently 1, 6 successfully completed

BSc: 4 successfully completed

Interns: 1 successfully completed, 1 prospect

RESEARCH GRANTS

"Advancing Cybersecurity Through Semantic Bugs in Interpreted Programming Languages" (PI) funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy - EXC 2092 CASA - 390781972 (approved from 2025 to 2028)

"Detecting Semantic Bugs in JavaScript Engines" unrestricted gift from Google Research Scholar ($60,000 USD)

"Cross-Layer Security Analysis, Testing, and Attestation for WebAssembly" (co-PI) funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy - EXC 2092 CASA - 390781972 (approved from 2026 to 2029)

PUBLICATIONS

Conference

C20 Zheng H., **Toffalini F.**, Liu Q., and Payer M.
"Squeezing Juicy Variant Bugs Out of Modern Browsers" Proceeding of the 20th USENIX WOOT Conference on Offensive Technologies (WOOT 2026) - **Best Paper**

C19 Sanft M., and **Toffalini F.**
"SemSan: A Configurable Sanitizer for Detecting Semantic Bugs" Proceeding of the 20th USENIX WOOT Conference on Offensive Technologies (WOOT 2026)

C18 Dechand S., Wienand T., Meumertzheim F., Samarin P., Resch S., Yakdan K., Holz T., and **Toffalini F.**
"Jazzer: Coverage-Guided Fuzzing for Semantic Vulnerabilities in the Java Ecosystem" Proceeding of the 47th IEEE Symposium on Security and Privacy (S&P 2026)

C17 Vanoverloop D., Sanchez A., **Toffalini F.**, Piessens F., Payer M., and Van Bulck J.
"TLBlur: Compiler-assisted automated hardening against controlled channels on off-the-shelf Intel SGX platforms" Proceeding of the 34nd USENIX Security Symposium (Usenix SEC 2025)

C16 Badoux N., **Toffalini F.**, and Payer M.
"Sourcerer: channeling the void" Proceeding of the 22th International Conference of Detection of Intrusions, Malware, and Vulnerability Assessment (DIMVA 2025)

C15 **Toffalini F.**, Badoux N., Tsinadze Z., and Payer M.
"Liberating libraries through automated fuzz driver generation: Striking a Balance Without Consumer Code" Proceeding of the ACM International Conference on the Foundations of Software Engineering (FSE 2025)

C14 Zheng H., **Toffalini F.**, Böhme M., and Payer M.
"MendelFuzz: The Return of the Deterministic Stage" Proceeding of the ACM International Conference on the Foundations of Software Engineering (FSE 2025)

C13 Wachter L., Gremminger J., Wressnegger C., Payer M., and **Toffalini F.**
"DUMPLING: Fine-grained Differential JavaScript Engine Fuzzing" Proceeding of the 32th Network and Distributed System Security Symposium (NDSS 2025) - **Distinguished Paper**

C12 Badoux N., **Toffalini F.**, Jeon Y., and Payer M.
"type++: Prohibiting Type Confusion With Inline Type Information" Proceeding of the 32th Network and Distributed System Security Symposium (NDSS 2025) - **Distinguished Paper**

C11 Srivastava P., **Toffalini F.**, Vorobyov K., Gauthier F., Bianchi A., and Payer M.
"Crystallizer: A Hybrid Path Analysis Framework To Aid in Uncovering Deserialization Vulnerabilities" Proceeding of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE 2023)

C10 Zheng H., Zhang J., Huang Y., Ren Z., Wang H., Cao C., Zhang Y., **Toffalini F.**, and Payer M.
"FishFuzz: Throwing Larger Nets to Catch Deeper Bugs" Proceeding of the 32nd USENIX Security Symposium (Usenix SEC 2023)

C9 Xu J., Di Bartolomeo L., **Toffalini F.**, Mao B., and Payer M.
"WarpAttack: Bypassing CFI through Compiler-Introduced Double-Fetches" Proceeding of the 44th IEEE Symposium on Security and Privacy (S&P 2023)

C8 Liu Q., **Toffalini F.**, Zhou Y., and Payer M.
"ViDeZZO: Dependency-aware Virtual Device Fuzzing" Proceeding of the 44th IEEE Symposium on Security and Privacy (S&P 2023)

C7 **Toffalini F.**, Payer M., Zhou J., and Cavallaro L.
"Designing a Provenance Analysis for SGX Enclaves" Proceeding of the 38th Annual Computer Security Applications Conference (ACSAC 2022)

C6 Jiang Z., Gan S., Herrera A., **Toffalini F.**, Romerio L., Tang C., Egele M., Zhang C., and Payer M.
"Evocatio: Conjuring Bug Capabilities from a Single PoC" Proceeding of the ACM SIGSAC Conference on Computer and Communications Security (CCS 2022)

C5 **Toffalini F.**, Graziano M., Conti M., and Zhou J.
"SnakeGX: a sneaky attack against SGX Enclaves" Proceeding of the 19th International Conference on Applied Cryptography and Network Security (ACNS 2022)

C4 **Toffalini F.**, Losiouk E., Biondo A., Zhou J., and Conti M.
"ScaRR: Scalable Runtime Remote Attestation for Complex Systems" Proceeding of the 22nd International Symposium on Research in Attacks, Intrusions and Defenses (RAID 2019)

C3 **Toffalini F.**, Ochoa M., Sun J., and Zhou J.
"Careful-Packing: A Practical and Scalable Anti-Tampering Software Protection enforced by Trusted Computing" Proceeding of the 9th ACM Conference on Data and Application Security and Privacy (CODASPY 2019)

C2 **Toffalini F.**, Sun J., and Ochoa M.
"Static Analysis of Context Leaks in Android Applications" Proceeding of the 40th International Conference on Software Engineering: Software Engineering in Practice (SEPA@ICSE)

C1 **Toffalini F.**, Abba' M., Carra D., and Balzarotti D.
"Google Dorks: Analysis, Creation, and new Defenses" Proceeding of the 13th International Conference of Detection of Intrusions, Malware, and Vulnerability Assessment, (DIMVA 2016)

Workshop

W8 Wienand T., Bernhard L., and **Toffalini F.**
"TYPEFUZZ: Type Coverage Directed JavaScript Engine Fuzzing (Registered Report)" Proceeding of 5th International Fuzzing Workshop (FUZZING) 2026

W7 Rusconi D., Yousef O., Picca M., Bruschi D., **Toffalini F.**, and Lanzi A.
"E-FuzzEdge: Efficient In-Place Firmware Fuzzing via Parallel Scheduling (Short Paper)" Proceeding of 5th International Fuzzing Workshop (FUZZING) 2026

W6 Rusconi D., Zoia M., Buccioli L., Pierazzi F., Bruschi D., Cavallaro L., **Toffalini F.**, and Lanzi A.
"EmbedWatch: Fat Pointer Solution for Detecting Spatial Memory Errors in Embedded Systems" Proceeding of the 6th Workshop on CPS and IoT Security (CPSIoTSec)

W5 Zheng H., **Toffalini F.**, and Payer M.
"TuneFuzz: Adaptively Exploring Target Programs" Proceeding of the 17th Intl. Workshop on Search-Based and Fuzz Testing (SBFT 2024)

W4 Visintin A., **Toffalini F.**, Losiouk E., Conti M., and Zhou J.
"HolA: Holistic and autonomous attestation for IoT networks" Proceeding of the Applied Cryptography and Network Security Workshops (ACNS 2022) - **Best Paper**

W3 **Toffalini F.**, Homoliak I., Harilal A., Binder A., and Ochoa M.
"Detection of Masqueraders Based on Graph Partitioning of File System Access Events" Proceeding of IEEE Security and Privacy Workshops (SPW)

W2 Harilal A., **Toffalini F.**, John C., Guarnizo J., Homoliak I., and Ochoa M.
"TWOS: A Dataset of Malicious Insider Threat Behavior Based on Gamified Competition" Proceeding of the 9th ACM CCS International Workshop on Managing Insider Security Threats (MIST)

W1 De Stefani F., Gamba P., Goldoni E., Savioli A., Silvestri D., and **Toffalini F.**
"REnvDB, a RESTful Database for Pervasive Environmental Wireless Sensor Networks" Proceeding of the 30th IEEE International Conference on Distributed Computing Systems Workshops

Journal

J5 Zoia M., Picca M., Rusconi D., Monzani A., **Toffalini F.**, Bruschi D., and Lanzi A.
"RemOTA: Remote attestation for detecting use-after-free in low-power microcontrollers" Computers Security, 2026

J4 **Toffalini F.**, Oliveri A., Graziano M., Zhou J., and Balzarotti D.
"The evidence beyond the wall: Memory forensics in SGX environments" Forensic Science International: Digital Investigation, 2021

J3 Homoliak I., **Toffalini F.**, Guarnizo J., Elovici Y., and Ochoa M.
"Insight Into Insiders and IT: A Survey of Insider Threat Taxonomies, Analysis, Modeling, and

Countermeasures" ACM Computing Surveys (CSUR), 2019

J2 **Toffalini F.**, Sun J., and Ochoa M.
"Practical static analysis of context leaks in Android applications" Software: Practice and Experience, 2019

J1 Harilal A., **Toffalini F.**, Homoliak I., John C., Guarnizo J., Mondal S., and Ochoa M.
"The Wolf Of SUTD (TWOS): A Dataset of Malicious Insider Threat Behavior Based on a Gamified Competition" Journal of Wireless Mobile Networks, Ubiquitous Computing, and Dependable Applications (JoWUA), 2018

ACADEMIC SERVICE

ACM CCS Doctoral Symposium 2026
ACM CCS reviewer 2026
NDSS reviewer 2022–2026
USENIX Sec reviewer 2025–2027
DIMVA reviewer 2022–2026
DIMVA poster chair 2024/25
ACSAC reviewer 2024–2026
ISSTA reviewer 2024/25
TOSEM reviewer 2024
Usenix SEC AE reviewer 2022
EuroSP shadow-reviewer 2020
TIFS reviewer 2018/19

COMMUNITY SERVICE

RUB's Doctoral Board
RUB's Discussion Group for 2026 reaccreditation of the cybersecurity program
PI in CASA 2.0
Member, CASA Early Career Researcher (ECR) Panel
Organizing CASA's CyBER series
Reviewer for DFG Sachbeihilfe Scientific Proposal
PhD Thesis Reviewer: Sapienza University of Rome, University of Milan, and RUB

INDUSTRY TALKS

"No App? No Problem: Fuzzing Libraries With Automatic Driver Generation" BlackAlps 2025

"SemSan: Finding the 0-Days You Can't Crash Into" BlackAlps 2026

OTHER ACTIVITIES

Employee in IT Companies: 2009-2011
Part-time Student during Master: 2012-2014